Problem K
Christmas lights

Time limit: **3 seconds**
Memory limit: **1024 megabytes**

Problem Description

Christmas is just around the corner, yet the team's progress remains stagnant. Before leaving work, the director hastily called for a meeting, trying to boost morale. On stage, his speech sounded like a hot-blooded anime character giving a rallying cry before a final battle. Jimmy, however, felt nothing but exhaustion.

On his way home, he passed through a park and saw rows of bare trees strung with old Christmas lights. The bulbs flickered weakly, their glow as fragile as an old man on his deathbed, yet they still struggled to shine-offering the last bit of warmth to the holiday, and to this cold, indifferent world.

But Jimmy also noticed that some of the lights had already gone completely dark. He wrote down the numbers of the broken bulbs, but he had no idea how to turn that information into a clear status map.

Now, your task is to help Jimmy solve this problem with a program.

The forest can be represented as an $N \times N$ grid, with the lights numbered in a spiral order traversal order (as shown below, $N = 3$):

$$\begin{matrix} 1 & 2 & 3 \\ 8 & 9 & 4 \\ 7 & 6 & 5 \end{matrix}$$

You need to generate a light status map of the forest, where `1` represents a working light and `0` represents a broken light.

Input Format

Your program will read input from standard input. The input begins with an integer T, the number of test cases. Each test case consists of two lines. The first line contains two integers, N and M, representing the size of the square forest and the number of broken bulbs. The second line contains M integers, $S_i (1 \leq i \leq M)$, which indicate the indices of the broken bulbs.

Output Format

Your program will write output to standard output. For each test case, print an $N \times N$ matrix where each entry represents the status of a bulb. Use '`1`' to indicate a working bulb and '`0`' to indicate a broken bulb. Each row of the matrix should be printed on a separate line, with entries separated by spaces. Please see the sample output.

Technical Specification

- $1 \le T \le 100$
- $1 \le N \le 10$
- $1 \le M \le N \times N$
- $1 \le S_i \le M, 1 \le i \le M$

Sample Input 1

```
2
3 2
2 7
4 3
6 11 15
```

Sample Output 1

```
1 0 1
1 1 1
0 1 1
1 1 1 1
1 1 1 1
0 1 0 0
1 1 1 1
```